

Moritz Akiya Zanger

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ABSTRACT

My name is Moritz Zanger and I am a final-year Ph.D. candidate at TU Delft. My doctoral research focuses on efficient uncertainty quantification for deep reinforcement learning. My main expertise includes deep learning, probabilistic modeling, uncertainty quantification methods, and reinforcement learning. More broadly, my goal is to advance machine learning algorithms that enable autonomous systems to reason under uncertainty, act reliably in complex environments, and ultimately benefit society.

EDUCATION

- **PH.D. CANDIDATE IN COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE** *Jun. 2021 - present*
Delft University of Technology, NL
Supervised by Prof. Matthijs T. J. Spaan and Dr. Wendelin Böhmer
Research Topic: Efficient Uncertainty Quantification in Deep Reinforcement Learning (working title).
EU-Horizon project Epistemic AI (epistemic-ai.eu).
- **M.SC. MECHANICAL ENGINEERING** *Oct. 2017-Nov. 2020*
Karlsruhe Institute of Technology, GER
Final Grade: 1.2 - with distinction (94th percentile in class)
Majors: Robotics · Medical Engineering
Final Thesis · Supervised by Prof. Marius J. Zöllner
Model-Based Reinforcement Learning for Constrained Policy Optimization in Robot Locomotion
- **B.SC. MECHANICAL ENGINEERING** *Oct. 2012 - Oct. 2017*
Karlsruhe Institute of Technology, GER
Final Grade: 1.6 (94th percentile in class)
Majors: Engineering Design

EXPERIENCE

- **Research Center of Information Technology (FZI)**
RESEARCH ASSISTANT - Dpt. Technical Cognitive Systems *Jan. 2021 - May 2021*
Gradient estimation for trust-region methods in model-based reinforcement learning.

RESEARCH ASSISTANT - Dpt. Intelligent Systems and Production Engineering *May 2019 - Dec. 2019*
Natural language processing with pretrained models (BERT) for requirements management.
- **TOHOKU UNIVERSITY, JP** *Sep. 2015-Sep. 2016*
RESEARCH ASSISTANT - Nagatani Field Robotics Lab
Robust LIDAR mappings for autonomous navigation in hazardous terrains.

PUBLICATIONS

- **Moritz A. Zanger**, Karam Daaboul, and J. Marius. Zöllner "Safe continuous control with constrained model-based policy optimization." 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS).
- **Moritz A. Zanger**, Wendelin Böhmer, and Matthijs T. J. Spaan. "Diverse Projection Ensembles for Distributional Reinforcement Learning" 2024 International Conference on Learning Representations (ICLR).
- Yaniv Oren, **Moritz A. Zanger**, Pascal R. van der Vaart, Matthijs T. J. Spaan and Wendelin Böhmer. "Value Improved Actor Critic Algorithms" 2025 Advances in Neural Information Processing Systems (NeurIPS).
- Max Weltevrede, **Moritz A. Zanger**, Matthijs T. J. Spaan and Wendelin Böhmer. "How Ensembles of Distilled Policies Improve Generalisation in Reinforcement Learning" 2025 Advances in Neural Information Processing Systems (NeurIPS).
- **Moritz A. Zanger**, Pascal R. van der Vaart, Matthijs T. J. Spaan and Wendelin Böhmer. "Contextual Similarity Distillation: Ensemble Uncertainties with a Single Model" 2026 International Conference on Learning Representations (ICLR).
- **Moritz A. Zanger**, Max Weltevrede, Yaniv Oren, Pascal R. Van der Vaart, Caroline Horsch, Wendelin Böhmer, Matthijs T. J. Spaan. "Universal Value-Function Uncertainties" 2026 International Conference on Learning Representations (ICLR).
- Guopeng Li, **Moritz A. Zanger**, Matthijs T. J. Spaan and Julian F. P. Kooij. "Cholesky Ordered Projection Q-learning (COP-Q): Guiding Safety-First Exploitation-Exploration by Multi-Objective Uncertainty" Under Review, 2026.

TEACHING

- CS4210 Intelligent Decision Making Project, Instructor TU Delft, NL, 2024 - present
- CSE3000 Bachelor Graduation Projects, Instructor: *Evaluating Robustness of Deep Reinforcement Learning for Autonomous Driving*. TU Delft, NL, 2023
- CS4400 Deep Reinforcement Learning: Lecture notes. TU Delft, NL, 2022 - present

SKILLS

TOOLS & FRAMEWORKS

Git • Jax • PyTorch • Tensorflow • Slurm • Singularity/Apptainer • ROS • MuJoCo • Unity • CARLA • PTC Creo

LANGUAGES

Python • Java • C++ • C# • SQL • MATLAB

SPOKEN LANGUAGES

German (fluent) • English (fluent, Toefl iBT 111/120) • Japanese (fluent) • Dutch(B1) • French(A1)

AWARDS & HONORS

- 2022** **MLSS Scholarship**, Machine Learning Summer School in Krakow, Poland.
- 2017-2020** **Students@Bosch Fellow**, Robert Bosch GmbH
- 2017** **GfSE Student Award**, Day of the Systems Engineering conference (TdSE) by the German Association for Systems Engineering (GfSE)
- 2015-2016** **DAAD Annual Scholarship**, German Academic Exchange Service (DAAD)

REFERENCES

Prof. Matthijs T. J. Spaan

Director • Sequential Decision Making • TU Delft
m.t.j.spaan@tudelft.nl

Dr. Wendelin Böhmer

Assistant Professor • Sequential Decision Making, TU Delft
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Prof. Frans A. Oliehoek

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